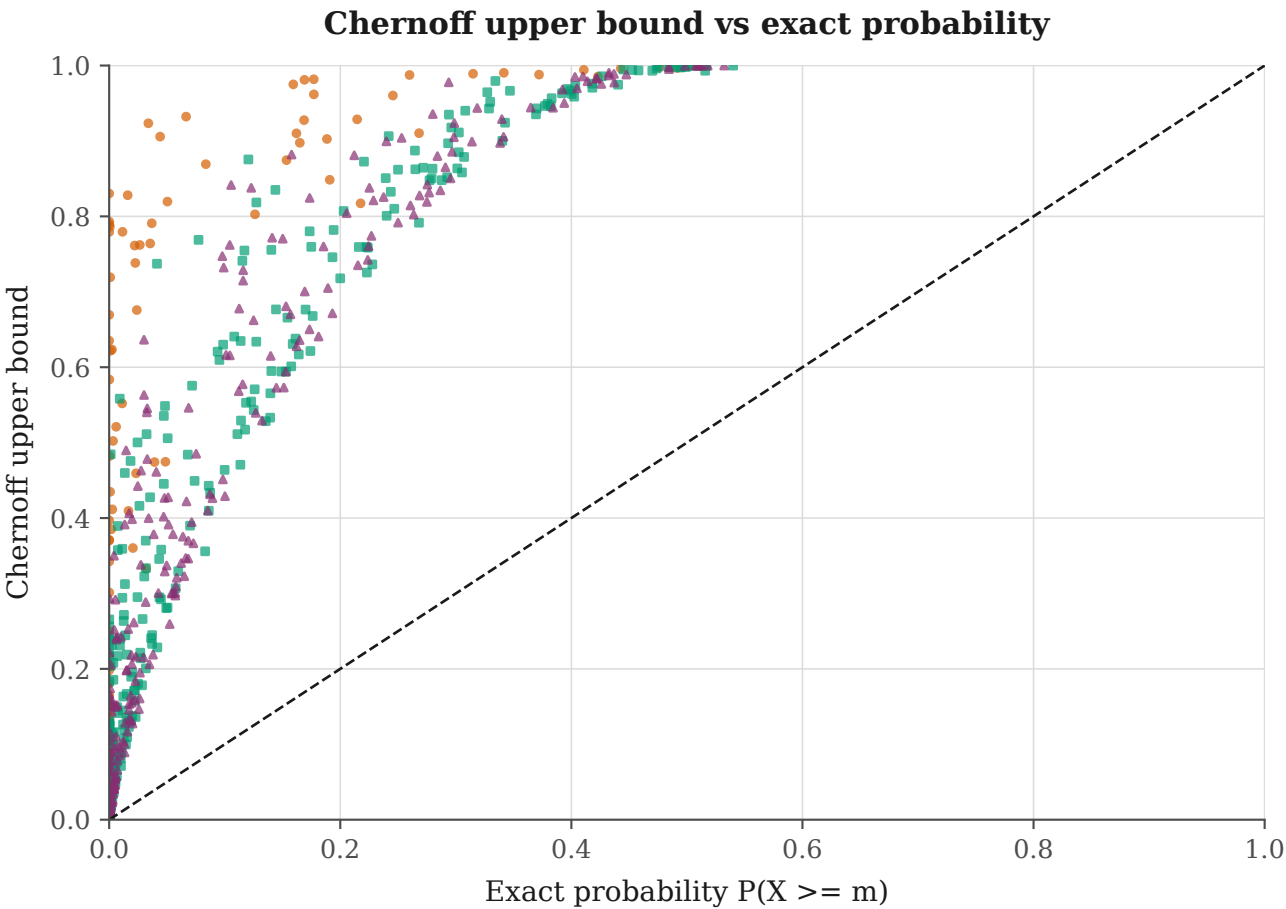

Chernoff upper bound vs exact probability
Chernoff upper bound
Exact probability P(X >= m)
0.0
0.2
0.4
0.6
0.8
1.0